



Exam :

End Term Quiz

Subject :

Maths2

Total Marks :

50.00

QP :

2025 Apr13: IIT M AN EXAM QIM4

Exam Mode

Learning Mode

View Question Paper Summary

QUESTION MENU

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TIMER

00:20



CONTROLS

SUBMIT EXAM

Your Score

0.00 / 50.00

(0%)

Question 1 : 6406531250640

Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT **"FOUNDATION LEVEL : MATHEMATICS FOR DATA SCIENCE II (COMPUTER BASED EXAM)"** ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT? CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN. (IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

OPTIONS :

☐ YES

☐ NO

Your score : 0

Discussions (0)



Question 2 : 6406531250641

View Solutions (1)

Total Mark : 1.00 | Type : MCQ

Find the set of all possible values for $\alpha \in \mathbb{R}$ such that the set $\{(1, \alpha, 0), (0, 2, -\alpha), (3, \alpha^2, \alpha)\}$ is linearly dependent.



OPTIONS :

☐ $\{0\}$

☐ $\{\alpha \in \mathbb{R} \mid \alpha^2 - 3\alpha + 2 = 0\}$

☐ $\{0, 1, 2\}$

☐ $\{\alpha \in \mathbb{R} \mid \alpha^2 + 3\alpha - 2 = 0\}$

Your score : 0

Discussions (0)



Question 3 : 6406531250642

View Solutions (0)

Total Mark : 2.00 | Type : MSQ

For which of the following functions defined on $\mathbb{R}^2 \setminus \{(0, 0)\}$ does the limit of the function exist at $(0, 0)$? 

OPTIONS :

☐ $f_1(x, y) = \sin\left(\frac{1}{x^2+y^2}\right)$ 

☐ $f_2(x, y) = (x^2 + y^2) \sin\left(\frac{1}{x^2+y^2}\right)$ 

☐ $f_3(x, y) = \frac{\sin(x^2+y^2)}{x^2+y^2}$ 

☐ $f_4(x, y) = \sin\left(\frac{x^2-y^2}{x^2+y^2}\right)$ 

Your score : 0

 Discussions (0)



Question 4 : 6406531250643

 View Solutions (0)

Total Mark : 2.00 | Type : MSQ

Which of the following sequences in $\mathbb{R}^2$ are convergent? 

OPTIONS :

☐ $\left\{ \left(\frac{\cos n}{n}, \frac{1}{n} \right) \right\}_{n \in \mathbb{N}}$ 

☐ $\left\{ \left(e^{\frac{1}{n}}, \frac{1}{e^n} \right) \right\}_{n \in \mathbb{N}}$ 

☐ $\left\{ \left(n \cos\left(\frac{1}{n}\right), \frac{(-1)^n}{n} \right) \right\}_{n \in \mathbb{N}}$ 

☐ $\left\{ \left((-1)^n, \frac{\sin n}{n} \right) \right\}_{n \in \mathbb{N}}$

Your score : 0

Discussions (1)



Question 5 : 6406531250644

View Solutions (1)

Total Mark : 3.00 | Type : MSQ

Which of the following options is/are true?

OPTIONS :

☐ If b is a linear combination of the columns of the coefficient matrix A , then the system of linear equations $Ax = b$ has a solution.

☐ Consider a system of linear equations $Ax = b$. If b is a linear combination of the columns of the coefficient matrix A , then the system $Ax = b$ has a unique solution.

☐ If A is an $m \times n$ matrix such that $m < n$, then $Ax = 0$ has a unique solution.

☐ If A is an $m \times n$ matrix such that $\text{rank}(A) = n$, then the system $Ax = 0$ will have a unique solution.

Your score : 0

Discussions (0)



Question 6 : 6406531250645

View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Let V be an inner product space and W be a subspace of V . If P is the projection of V on W , choose the correct option(s).

OPTIONS :

☐ If $Pv = v$ for some $v \in V$, then $v \in W$.

☐ $\dim(W) = \dim(W^\perp)$. 

☐ If A is the matrix representation of P with respect to the same basis for both domain and co-domain, then $A^2 = A$. 

☐ Suppose $v \in W \cap W^\perp$, then $v = 0$ 

Your score : 0

 Discussions (0)



Question 7 : 6406531250646

 View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a scalar-valued multivariable function ($n \geq 2$) such that the partial derivatives exist at all points and are continuous. Choose the correct statements from the following. 

OPTIONS :

☐ For $x \in \mathbb{R}^n$, if the directional derivatives at x in all directions are zero, then x is a critical point of f . 

☐ The second order partial derivatives satisfy the relation 

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i},$$
 for all i, j , whenever they exist.

☐ The i^{th} component of ∇f gives the partial derivative of f with respect to the i^{th} variable x_i . 

☐ For each point, there exists a unit direction u such that the directional derivative of f at that point in the direction u is zero. 

Your score : 0

 Discussions (0)



Question 8 : 6406531250647

[View Solutions \(1\)](#) Total Mark : 1.00 | Type : SA

Let $A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$ and $\det(A) = 5$. Let $B = \begin{bmatrix} 3a_1 & -6b_1 & 3c_1 \\ a_2 & -2b_2 & c_2 \\ a_3 & -2b_3 & c_3 \end{bmatrix}$. Find $\det(B)$.

Answer (Numeric):

Accepted Answer : -30

Your score : 0

[Discussions \(0\)](#)


Question 9 : 6406531250648

[View Solutions \(1\)](#) Total Mark : 1.00 | Type : SA

Let $A = \begin{bmatrix} 7 & 3 \\ 4 & 3 \end{bmatrix}$. Find the determinant of the matrix $A^6 - A^5$.

Answer (Numeric):

Accepted Answer : 0

Your score : 0

[Discussions \(0\)](#)


Question 10 : 6406531250649

[View Solutions \(0\)](#) Total Mark : 1.00 | Type : SA

Let $A, B \in M_{4 \times 4}(\mathbb{R})$ such that $A^2 = 2I$ and $B^2 = 3I$. If $\det(A) < 0$ and $\det(B) > 0$, find $\det(AB)$.

Answer (Numeric):

Accepted Answer : -36

Your score : 0

 Discussions (0)**Question 11 : 6406531250650** View Solutions (1)

Total Mark : 2.00 | Type : SA

If (a, b) is the projection of the vector $(3, 1)$ on the line $y = 2x$ using the standard inner product, find $a + b$.



Answer (Numeric):

Accepted Answer : 3

Your score : 0

 Discussions (0)**Question 12 : 6406531250651**

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Let $v_1 = (1, -1, 1), v_2 = (2, 0, 4), v_3 = (1, 1, t)$.



Your score : 0

**Question 13 :
6406531250652** View Parent QN View Solutions (1)

Total Mark : 1.00 | Type : SA

Find the value of $t \in \mathbb{R}$ such that
 $v_3 \in \text{span}\{v_1, v_2\}$.



Answer (Numeric):

Answer

Accepted Answer : 3

Your score : 0

Discussions (0)



Question 14 :
6406531250653



View Parent QN



View Solutions (0)

Total Mark : 1.00 | Type : SA

Let v_1, v_2, v_3 be as in the main question
with the value of t obtained.
If $v_3 = \alpha v_1 + \beta v_2$, find $\alpha - \beta$.



Answer (Numeric):

Answer

Accepted Answer : -2

Your score : 0

Discussions (0)



Question 15 : 6406531250654

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation such that $T(1, 1) = (4, 5)$
and $T(1, 0) = (3, -2)$.



Your score : 0



$$T(x, y) = yT(1, 1) + x-yT(1, 0) \\ y(1, 5) + x-y(3, -2)$$

D14

Question 16 :
6406531250655

View Parent QN

View Solutions (1)

Total Mark : 1.00 | Type : MCQ

Find the expression for $T(x, y)$.



OPTIONS :

☐ $T(x, y) = (x + 3y, 7x - 2y)$

☐ $T(x, y) = (x - 3y, -7x + 2y)$

☒ $T(x, y) = (3x + y, -2x + 7y)$

☐ $T(x, y) = (3x - y, 2x + 7y)$

$$\begin{bmatrix} 3 & 1 \\ -2 & 7 \end{bmatrix}$$

Your score : 0

Discussions (0)



Question 17 :
6406531250656

View Parent QN

View Solutions (0)

Total Mark : 1.00 | Type : MCQ

Suppose A is the matrix representation of T with respect to the standard basis for both domain and co-domain. Then A is given by

OPTIONS :

☐ $\begin{bmatrix} 3 & -2 \\ 1 & 7 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & 3 \\ -7 & 2 \end{bmatrix}$

☐ $\begin{bmatrix} 3 & 1 \\ -2 & 7 \end{bmatrix}$ 

☐ $\begin{bmatrix} 3 & -1 \\ 2 & -7 \end{bmatrix}$ 

Your score : 0

 Discussions (0)



Question 18 : 6406531250657

Total Mark : 0.00 | Type : COMPREHENSION

Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a linear transformation and A be a matrix such that $T(x) = Ax$ for all $x \in \mathbb{R}^3$. Answer the given subquestions. 

Your score : 0



Question 19 : 6406531250658

 View Parent QN

 View Solutions (1)

Total Mark : 1.00 | Type : SA

What is the maximum rank of A ?

Answer (Numeric):

Answer

Accepted Answer : 2

Your score : 0

 Discussions (0)



Question 20 :
6406531250659[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : SA

What is the minimum value for nullity of A?

Answer (Numeric):

Answer

Accepted Answer : 1

Your score : 0

[Discussions \(0\)](#)**Question 21 :**
6406531250660[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : MCQ

Choose the correct option for T.

OPTIONS :

- ☐ T may be one-one but not onto.
- ☐ T may be onto but not one-one.
- ☐ T must be both one-one and onto.
- ☐ T can neither be one-one nor onto.

Your score : 0

[Discussions \(0\)](#)**Question 22 : 6406531250661**

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Consider the function f defined by $f(x, y) = \ln(x^2 + y^2 - 1)$ on an appropriate domain in $\mathbb{R}^2$.



Your score : 0



Question 23 :
6406531250662



View Parent QN



View Solutions (0)

Total Mark : 1.00 | Type : MCQ

What is the minimum number of points required to be removed from $\mathbb{R}^2$ to obtain a valid domain for f ? 

OPTIONS :

☐ 0☐ 1☐ Infinitely many

Your score : 0

 Discussions (0)

Question 24 :
6406531250663



View Parent QN



View Solutions (0)

Total Mark : 1.00 | Type : MCQ

What is the range of the function f when defined on the domain $D = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \geq 2\}$? 

OPTIONS :

☐ $(\infty, 0]$ ☐ $[0, \infty)$ ☐ $(-1, 0]$ 

☐ $[0, 1)$ 

Your score : 0

 Discussions (0)



Question 25 :
6406531250664

 View Parent QN

 View Solutions (0)

Total Mark : 1.00 | Type : MCQ

The function f defined on the domain 
specified in the previous subquestion
is injective.

OPTIONS :

☐ TRUE

☐ FALSE

Your score : 0

 Discussions (0)



Question 26 : 6406531250665

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Consider the function $f(x, y) = \sin\left(\frac{x+y}{x-y}\right)$ defined on $\{(x, y) \in \mathbb{R}^2 \mid x < y\}$. 

Let (a, b) denote the unit direction of steepest descent for the function f at the point $(-1, 1)$.

Your score : 0



Question 27 :
6406531250666

[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : SA

Find the value of $\sqrt{2}a$. 

Answer (Numeric):

Accepted Answer : 1

Your score : 0

[Discussions \(0\)](#)

Question 28 :
6406531250667

[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : SA

Find the value of $\sqrt{2}b$. 

Answer (Numeric):

Accepted Answer : 1

Your score : 0

[Discussions \(0\)](#)

Question 29 :
6406531250668

[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : SA

If (c, d) denotes a direction vector along which there is no change in f at $(-1, 1)$. Find the value of $c + d$. 

Answer (Numeric):

Accepted Answer : 0

Your score : 0

 Discussions (0)

Question 30 : 6406531250669

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Consider the function $f(x, y) = x \sin y + y \cos x$ defined on $\mathbb{R}^2$.

Your score : 0

Question 31 :
6406531250670

View Parent QN



View Solutions (0)

Total Mark : 1.00 | Type : SA

If the directional derivative of f at the point $(0, \frac{\pi}{2})$ in the direction $(t, \sqrt{1-t^2})$ is equal to 0, then find the value of t^2 .



Answer (Numeric):

Accepted Answer : 0.5

Your score : 0

 Discussions (0)

Question 32 :
6406531250671 View Parent QN View Solutions (0)

Total Mark : 2.00 | Type : SA

Suppose $(1, b, c)$ is a point on the tangent line at $(0, \frac{\pi}{2})$ in the direction obtained in the previous part. Find the value of $b - c$.



Answer (Numeric):

Answer

Accepted Answer : -1

Your score : 0

 Discussions (0)**Question 33 : 6406531250672**

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Consider the function f defined on $\mathbb{R}^2$ by $f(x, y) = x^2 + y^2 - x^2y^2$.



Your score : 0

**Question 34 :**
6406531250673 View Parent QN View Solutions (0)

Total Mark : 2.00 | Type : MSQ

Which of the following are critical points of f ?



OPTIONS :

☐ (1, 0)☐ (0, 1)

☐ $(-1, 1)$ ☐ $(1, -1)$

Your score : 0

[Discussions \(0\)](#)**Question 35 :**
6406531250674[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : SA

It can be checked that $(0, 0)$ is a critical point of f . Find the determinant of the Hessian matrix of f at $(0, 0)$.



Answer (Numeric):

Answer

Accepted Answer : 4

Your score : 0

[Discussions \(0\)](#)**Question 36 :**
6406531250675[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : MCQ

What can be concluded about the critical point $(0, 0)$ from the Hessian test?

OPTIONS :

☐ The Hessian test is inconclusive.☐ $(0, 0)$ is a local minimum.☐ $(0, 0)$ is a local maximum.☐ $(0, 0)$ is a saddle point.

Your score : 0

Discussions (0)



Question 37 : 6406531250676

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Consider the graph of a scalar-valued function $f(x, y)$ defined on a closed and bounded domain D in $\mathbb{R}^2$. The points on the graph that are closest to and farthest from a fixed point (x_0, y_0, z_0) can be found by computing the global extrema of the function defined by computing the square of the distance between (x_0, y_0, z_0) and a generic point $(x, y, f(x, y))$ on the graph. Note that the domain of the distance function is also D . For the given subparts, consider the function $f(x, y) = -2 - x^2 - y^2$ restricted to the domain $D = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$.



Your score : 0

Question 38 :
6406531250677

View Parent QN

View Solutions (0)

Total Mark : 1.00 | Type : MCQ

Using the procedure described in the main data, choose the expression for the function ϕ that measures the square of the distance between $(0, 0, 0)$ and a generic point on the graph of f .



OPTIONS :

☐ $\phi(x, y) = x^2 - y^2 - (x^2 + y^2 + 2)^2$



☐ $\phi(x, y) = x^2 + y^2 + (x^2 - y^2 - 2)^2$



☐ $\phi(x, y) = x^2 + y^2 + (x^2 + y^2 - 2)^2$ 

☐ $\phi(x, y) = x^2 + y^2 + (x^2 + y^2 + 2)^2$ 

Your score : 0

 Discussions (0)



Question 39 :
6406531250678

 View Parent QN

 View Solutions (0)

Total Mark : 1.00 | Type : MCQ

Find the number of critical points of the function ϕ obtained in the previous subquestion that lie in $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 < 1\}$, the interior of D .



OPTIONS :

- ☐ 0
- ☐ 1
- ☐ 2
- ☐ Infinitely many

Your score : 0

 Discussions (0)



Question 40 :
6406531250679

 View Parent QN

 View Solutions (0)

Total Mark : 2.00 | Type : MCQ

Choose the correct statement from the following.

OPTIONS :

- ☐ ϕ has finitely many global minima and finitely many global maxima in D .
- ☐ ϕ has finitely many global minima and infinitely many global maxima in D .
- ☐ ϕ has infinitely many global minima and finitely many global maxima in D .
- ☐ ϕ has infinitely many global minima and infinitely many global maxima in D .

Your score : 0

Discussions (0)



Question 41 :
6406531250680

View Parent QN

View Solutions (0)

Total Mark : 1.00 | Type : SA

Using the function ϕ , compute the distance between $(0, 0, 0)$ and a point on the graph of $f(x, y) = -2 - x^2 - y^2$ that is closest to $(0, 0, 0)$.

Answer (Numeric):

Answer

Accepted Answer : 2

Your score : 0

Discussions (0)



Question 42 : 6406531250681

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Let $u_1 = (-1, 1, -1)$, $u_2 = (2, 0, 2)$, $u_3 = (1, -2, 1)$ and $W = \text{span}\{u_1, u_2, u_3\}$.

Your score : 0



Question 43 :
6406531250682



View Parent QN



View Solutions (0)

Total Mark : 1.00 | Type : SA

Find the dimension of W.

Answer (Numeric):

Answer

Accepted Answer : 2

Your score : 0

Discussions (0)



Question 44 :
6406531250683



View Parent QN



View Solutions (0)

Total Mark : 1.00 | Type : MCQ

Let $U = \{(x, y, z) \mid x + y + z = 0\}$. Which of the following sets represents $W \cap U$?

OPTIONS :

☐ $\text{span}\{(-1, 0, 1)\}$

☐ $\text{span}\{(1, 2, 1)\}$

☐ $\{(x, y, z) \mid 2x = -y = 2z\}$

☐ $\{(x, y, z) \mid x = -2y = z\}$

Your score : 0

Discussions (0)



Question 45 :

6406531250684



View Parent QN



View Solutions (0)

Total Mark : 1.00 | Type : MCQ

Let A be a 3×3 matrix such that u_1 and u_2 are the first 2 columns of A . Choose the correct option from the following.



OPTIONS :

☐ If u_3 is the third column of A , then $Ax = 0$ has a unique solution.



☐ If $u_1 + u_2$ is the third column of A , then $Ax = 0$ has infinitely many solutions.



☐ The system $Ax = 0$ will not have any solution if the third column of A is u_1 or u_2 .



☐ If u_3 is the third column of A , then $Ax = b$ will not have a solution for any $b \in \mathbb{R}^3$.



Your score : 0

Discussions (0)



Question 46 :

6406531250685



View Parent QN



View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Which of the following vector spaces is/are isomorphic to W ?

OPTIONS :

☐ $W_1 = \mathbb{R}^2$ 

☐ $W_2 = \{x \in \mathbb{R}^3 \mid Ax = 0 \text{ where the columns of } A \text{ are } u_1, u_2 \text{ and } u_3\}$ 

☐ $W_3 = \{(x, y, z) \mid x + y - z = 0, x - y = 0\}$ 

☐ $W_4 = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mid a + c = 0, b + d = 0 \right\}$ 

Your score : 0

 Discussions (0)



 **SUBMIT EXAM**